Demand forecast & stock plan: summary

The question

How many units of each of our 200 best-selling products will we sell in the next 8 weeks, and how much stock do we need so we don't run out, including over Christmas?

Key results

15% less
forecast error than the best simple method (whole store, 4 weeks ahead: 12.0% vs 14.2%)

84%
of real weeks fell inside the forecast's low-high range (target 80%), tested on unseen weeks

177,238
units of stock recommended for the next 4 weeks, covering demand 9 times out of 10

What we found

- The combined forecast (machine learning + recent average) beat all 5 simple methods tested, for single products (39.4% error vs 42.0%) and for the whole store.
- Single-product weeks stay hard to predict: a few wholesale customers order thousands of units at once. That is why the stock plan uses 4-week totals and a safety range, not one number per week.
- Short weeks (Easter, bank holidays) and the Christmas closure are built into the forecast, so they are not mistaken for a fall in demand.
- Recommendation: order to the 'Stock to cover 4 weeks' column, and review products whose recent sales fall outside the forecast range.

Data and method

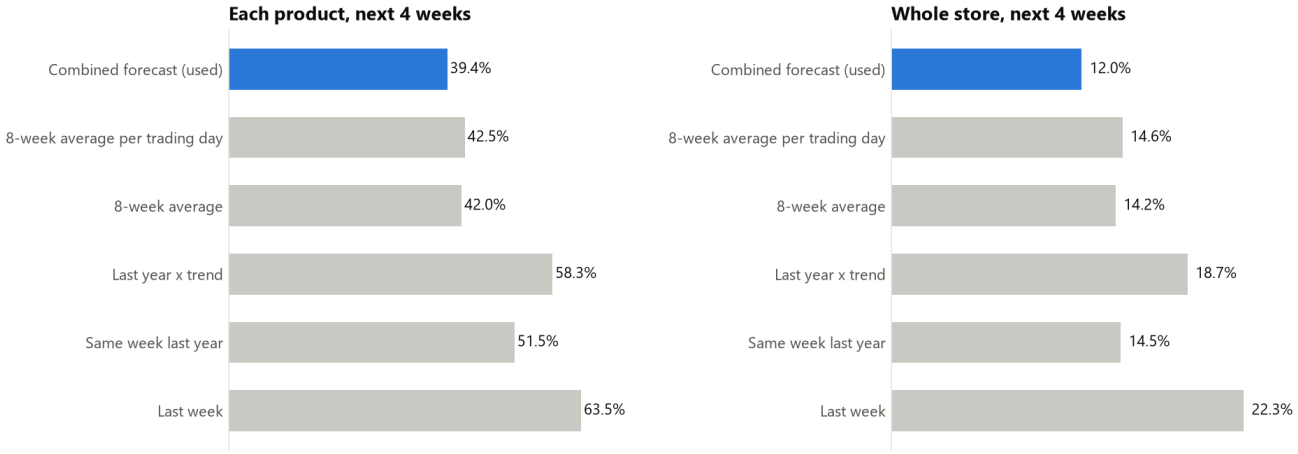
105 weeks of real sales (1.07M order lines, Dec 2009 to Dec 2011), cleaned in SQL. Weekly units for the 200 best sellers (39% of revenue). LightGBM models (expected value and low / high range) combined 50/50 with the 8-week average per trading day. Tested on 7 forecast dates in 2011, each using only earlier data.



How accurate is the forecast?

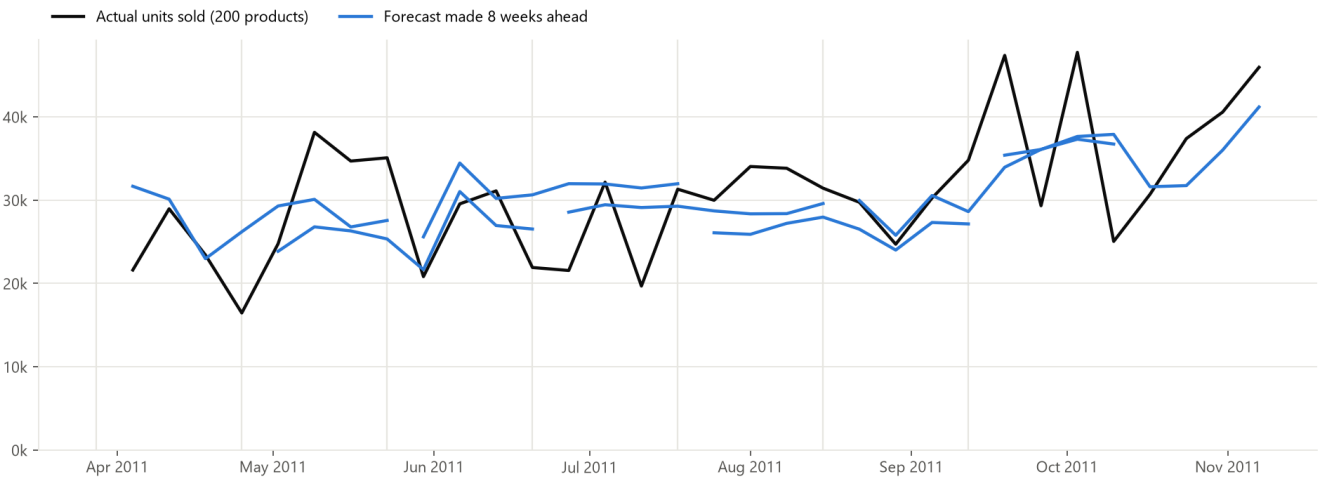
On 7 dates in 2011 the forecast was made using only the weeks before that date, then compared with what really sold over the next 8 weeks (9,694 product-weeks). Error = total difference between forecast and actual units / actual units (lower is better).

Forecast error (lower is better), tested on 7 dates in 2011



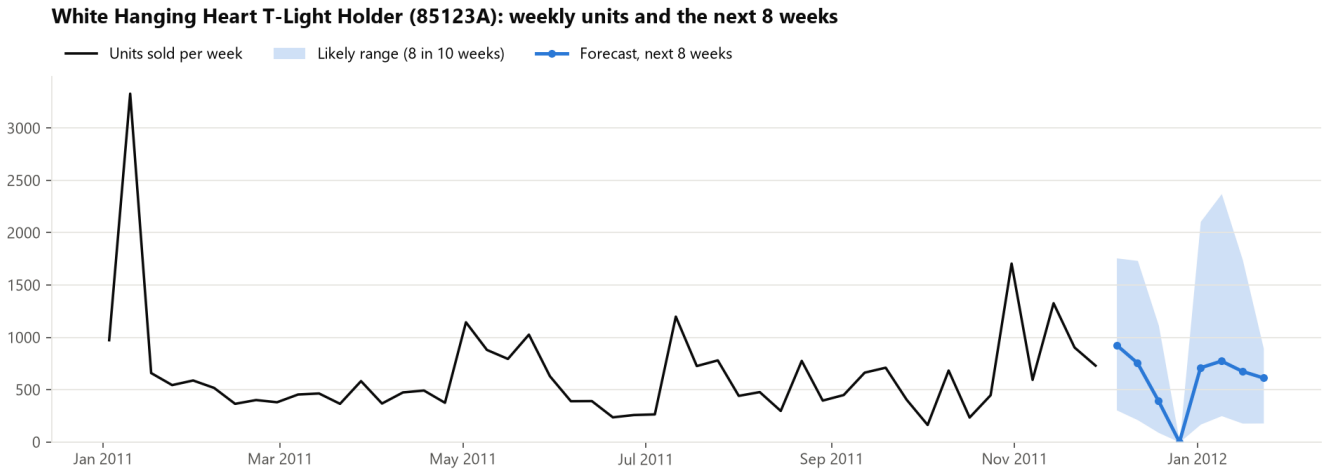
Method	Each product, 4 wk	Whole store, 4 wk	Bias
Combined forecast (used)	**39.4%**	**12.0%**	**-2.9%**
8-week average	42.0%	14.2%	-7.8%
Same week last year	51.5%	14.5%	+11.2%
Last year x trend	58.3%	18.7%	+15.7%
Last week	63.5%	22.3%	-2.2%

Weekly units: forecasts made on 7 dates in 2011 vs what really sold



Forecast and stock plan

Example product: weekly units sold in 2011, and the forecast for the next 8 weeks with its likely range (8 weeks out of 10 fall inside). The dip is the Christmas closure.



Stock plan: top-selling products (by revenue)

Product	Sold per week	Demand, next 4 wk	Stock for 4 wk (90%)
Regency Cakestand 3 Tier	222	596	**988**
White Hanging Heart T-Light Holder	828	2,074	**3,547**
Party Bunting	176	370	**629**
Jumbo Bag Red Retrospot	1,095	2,447	**3,318**
Paper Chain Kit 50'S Christmas	1,340	3,005	**4,887**
Assorted Colour Bird Ornament	1,002	2,520	**3,991**
Chilli Lights	375	1,022	**2,004**
Small Popcorn Holder	2,380	5,104	**8,616**
Jumbo Bag Pink Polkadot	394	930	**1,436**
Set Of 3 Cake Tins Pantry Design	135	362	**811**

Sold per week = average of the last 8 weeks. The Excel file the client gets (below) has all 200 products, the weekly forecast with low / high numbers, and the accuracy test.

Stock code	Product	Category	Sold per week, last 8 weeks	Expected demand, next 4 weeks	Stock to cover 4 weeks (90% of the time)	Expected demand, next 8 weeks	Revenue last 52 weeks (£)
22423	REGENCY CAKESTAND 3 TIER	Kitchen & dining	222	596	988	1,277	171,440
85123A	WHITE HANGING HEART T-LIGHT HOLDER	Candles & lighting	828	2,074	3,547	4,844	101,649
47566	PARTY BUNTING	Party & celebrations	176	370	629	1,013	99,203
85099B	JUMBO BAG RED RETROSPOT	Bags	1,095	2,447	3,318	5,922	93,823
22086	PAPER CHAIN KIT 50'S CHRISTMAS	Christmas	1,340	3,005	4,887	6,002	60,659
84879	ASSORTED COLOUR BIRD ORNAMENT	Garden	1,002	2,520	3,991	5,342	58,289
79321	CHILLI LIGHTS	Candles & lighting	375	1,022	2,004	2,162	53,564
22502	PICNIC BASKET WICKER SMALL	Home decor	0	11	40	19	51,380
22197	SMALL POPCORN HOLDER	Party & celebrations	2,380	5,104	8,616	11,404	47,506
22386	JUMBO BAG PINK POLKADOT	Bags	394	930	1,436	2,292	41,713
22720	SET OF 3 CAKE TINS PANTRY DESIGN	Kitchen & dining	135	362	811	817	37,222
22960	JAM MAKING SET WITH JARS	Kitchen & dining	200	487	825	1,125	37,124
21137	BLACK RECORD COVER FRAME	Home decor	326	864	1,921	1,814	35,377
20725	LUNCH BAG RED RETROSPOT	Bags	395	881	1,404	2,142	35,255
82484	WOOD BLACK BOARD ANT WHITE FINISH	Home decor	82	216	446	466	34,144
22469	HEART OF WICKER SMALL	Home decor	444	1,099	1,947	2,487	32,451
85099F	JUMBO BAG STRAWBERRY	Bags	215	501	736	1,289	32,377
22178	VICTORIAN GLASS HANGING T-LIGHT	Candles & lighting	497	1,251	2,056	2,789	31,979